

Customer Requirement Document (CRD): GaugeMaster

Date: April 20, 2026
Subject: Calibration Tracking System Enhancements & Automation

1. Objective

The goal is to enhance the existing GaugeMaster software with a multi-location calibration tracking system. This system will automate reminders to location heads, centralize digital certificate storage, and provide comprehensive history tracking for every instrument.

2. Multi-Location & Notification Logic

The system must support multiple company locations, each mapped to a specific Email ID (Location Head). The following status-driven notification logic will be implemented:

Status	Action / Trigger	Email Logic
15 Days for Due	Instrument is within 15 days of its calibration due date.	Daily reminder sent to the Location Head to submit the instrument to the lab.
Overdue	Instrument has passed its calibration due date.	Daily reminder sent to the Location Head until status changes.
Sent for Calibration	Instrument Sent for Calibration	All automated reminders stop immediately for this instrument.
Calibrated	Calibration process is complete.	Final notification to the Location Head to collect the instrument from the lab.

3. Digital Asset & History Management

- **Continuous Certificate Upload:** Users can upload a new calibration certificate every time an instrument returns from the lab.

- **Digital Archive:** The system must retain all old certificates. Users can view the entire history of certificates by opening the instrument's upload feature.
- **History Card Feature:** A dedicated "History Card" module to view and download the full calibration history (dates, results, etc.) for each instrument in the customer's specified format.

4. Reporting & Formatting Requirements

- **Bulk Upload Update:** Modify the bulk instrument upload tool to include extra columns as per the customer's updated format.
- **Customized Downloads:** Ability to download reports based on:
 - Date Range (Monthly, Quarterly, Yearly)
 - Instrument Status
 - Specific Location
- **Branding:** All generated PDF/Excel reports must feature the **Goodluck** logo and company name.
- **Automated Summary:** Regular reports sent to management email IDs summarizing the count of "Due," "Ongoing Calibration," and "Completed Calibration" items.

5. Technical Scope for Developers

- Database update to support Location-to-Email mapping.
- Implementation of a background scheduler (Cron job) for daily email triggers.
- Branding integration into the reporting engine.
- Adjustment of the CSV/Excel parser for bulk uploads.